

DATA ANALYTICS PROFESSIONAL +AI



Introduction To Python

- Why Python
- Application areas of python
- Python implementations
 1. Cpython
 2. Jython
 3. Ironpython
 4. Pypy
- Python versions
- Installing python
- Python interpreter architecture
- 5. Python byte code compiler
- 6. Python virtual machine(pvm)

Writing and Executing First Python Program

- Using interactive mode
- Using script mode
 1. General text editor and command window
 2. Idle editor and idle shell
- Understanding print() function
- How to compile python program explicitly

Python Language Fundamentals

- Character set
- Keywords
- Comments
- Variables
- Literals
- Operators
- Reading input from console
- Parsing string to int, float

Python Conditional Statements

- If statement
- If else statement
- If elif statement
- If elif else statement
- Nested if statement

Looping Statements

- While loop

- For loop
- Nested loops
- Pass, break and continue keywords

Standard Data Types

- Int, float, complex, bool, nonetype
- Str, list, tuple, range
- Dict, set, frozenset

String Handling

- What is string
- String representations
- Unicode string
- String functions, methods
- String indexing and slicing
- String formatting

Python List

- Creating and accessing lists
- Indexing and slicing lists
- List methods
- Nested lists
- List comprehension

Python Tuple

- Creating tuple
- Accessing tuple
- Immutability of tuple

Python Set

- How to create a set
- Iteration over sets
- Python set methods
- Python frozenset

Python Dictionary

- Creating a dictionary
- Dictionary methods
- Accessing values from dictionary
- Updating dictionary
- Iterating dictionary
- Dictionary comprehension

Python Functions

- Defining a function
- Calling a function
- Types of functions
- Function arguments
 1. Positional arguments, keyword arguments
 2. Default arguments, non-default arguments
 3. Arbitrary arguments, keyword arbitrary arguments

- Function return statement
- Nested function
- Function as argument
- Function as return statement
- Decorator function
- Closure
- Map(), filter(), reduce(), any() functions
- Anonymous or lambda function

Modules & Packages

- Why modules
- Script v/s module
- Importing module
- Standard v/s third party modules
- Why packages
- Understanding pip utility

File I/O

- Introduction to file handling
- File modes
- Functions and methods related to file handling
- Understanding with block

Regular Expressions(Regex)

- Need of regular expressions
- Re module
- Functions /methods related to regex
- Meta characters & special sequences

Object Oriented Programming

- Procedural v/s Object Oriented Programming
- OOP Principles
- Defining a Class & Object Creation
- Inheritance
- Encapsulation
- Polymorphism
- Abstraction
- Garbage Collection
- Iterator & Generator

Exception Handling

- Difference Between Syntax Errors and Exceptions
- Keywords used in Exception Handling
- try , except , finally , raise , assert
- Types of Except Blocks
- User-defined Exceptions

GUI Programming

- Introduction to Tkinter Programming
- Tkinter Widgets

- Layout Managers
- Event handling
- Displaying image

Multi-Threading Programming

- Multi-processing v/s Multi-threading
- Need of threads
- Creating child threads
- Functions /methods related to threads
- Thread synchronization and locking

SQL

Introduction to RDBMS

- What is Relational Database Package
- Difference between SQL & Database
- Installing MySQL Server database

SQL Basic

- DDL: Create, Alter, Drop, etc.
- DML: Insert, Update, Delete ,etc.
- DQL : Select
- Autoincrement field
- SQL Comments
- SQL Aliases
- Savepoint & rollback

SQL Constraints

- Not NULL, Unique key
- Primary key, Check
- Default, Foreign key

SQL Operators

- Arithmetic operators
- Logical operators
- Conditional operators
- Like, between, in operators

SQL Clauses

- Order by
- Where
- Limit/top
- Group by
- having

SQL Joins

- Inner Join
- Left Join
- Right Join
- Full Join

SQL View

- creating view

- updating view
- fetching data from view

SQL Functions

- String functions
- Aggregate functions
- Date & time functions

Stored Procedures & Functions

- Understanding stored procedures and their key benefits
- Working with stored procedures
- Studying user-defined functions

Working with CSV Files:

- How to write result to csv file
- How to read csv file

Python Database Connectivity

- Database Drivers and connectors
- Creating connection object
- Understanding cursor object
- Executing SQL statements using cursor
- Fetching records from cursor
- Storing and retrieving Date and Time

MongoDB

Introduction To MongoDB

- Understanding NoSQL DB
- NoSQL vs. SQL DB
- Understanding Mongo DB
- Downloading & Installation
- Introduction of MongoDB shell and Compass
- Understanding database, collection & document

Crud Operations

- Insert Document
- Delete Document
- Update Document
- Query Document

Operators In MongoDB

- Query and Projection operators
- Update operator
- Aggregation Pipeline operators

Methods In MongoDB

- limit and sort
- bulk methods
- other methods

Indexing And Relationships

- Types of Indexes
- Creating an Indexes
- Dropping an Indexes
- Defining Relationships between Documents

Python Connectivity With MongoDB

- Introduction to pymongo
- Installing pymongo module
- MongoClient
- Getting database and collection
- CRUD operations
- Range Queries

Statistics & Analytics

Numpy Package

- Difference between list and numpy array
- Vector and matrix operations
- Array indexing and slicing

Pandas Package

Introduction to pandas

- Labeled and structured data
- Series and dataframe objects

How to load datasets

- From excel
- From csv
- From html table

Accessing data from Data Frame

- at & iat
- loc & iloc
- head() & tail()

Exploratory Data Analysis (EDA)

- describe()
- groupby()
- crosstab()
- boolean slicing / query()

Data Manipulation & Cleaning

- Map(), apply()
- Combining data frames
- Adding/removing rows & columns
- Sorting data
- Handling missing values
- Handling duplicacy
- Handling data error

Handling Date and Time

Data Visualization using matplotlib and seaborn packages

- Scatter plot, lineplot, bar plot
- Histogram, pie chart,
- Jointplot, pairplot, heatmap
- Outlier detection using boxplot

Advanced Excel

Advanced Excel Course - Overview of the Basics of Excel

- Customizing common options in Excel
- Absolute and relative cells
- Protecting and un-protecting worksheets and cells

Advanced Excel Course - Working with Functions

- Writing conditional expressions (using IF)
- Using logical functions (AND, OR, NOT)
- Using lookup and reference functions (VLOOKUP, HLOOKUP, MATCH, INDEX)
- Vlookup with Exact Match, Approximate Match
- Nested Vlookup with Exact Match
- Vlookup with Tables, Dynamic Ranges
- Nested Vlookup with Exact Match
- Using VLOOKUP to consolidate Data from Multiple Sheets

Advanced Excel

Advanced Excel Course - Data Validations

- Specifying a valid range of values for a cell
- Specifying a list of valid values for a cell
- Specifying custom validations based on formula for a cell

Advanced Excel Course - Working with Templates

- Designing the structure of a template
- Using templates for standardization of worksheets

Advanced Excel Course - Sorting and Filtering Data

- Sorting tables
- Using multiple-level sorting
- Using custom sorting
- Filtering data for selected view (AutoFilter)
- Using advanced filter options

Advanced Excel Course - Working with Reports

- Creating subtotals
- Multiple-level subtotals
- Creating Pivot tables
- Formatting and customizing Pivot tables
- Using advanced options of Pivot tables
- Pivot charts
- Consolidating data from multiple sheets and files using Pivot tables
- Using external data sources

- Using data consolidation feature to consolidate data
- Show Value As (% of Row, % of Column, Running Total, Compare with Specific Field)
- Viewing Subtotal under Pivot
- Creating Slicers (Version 2010 & Above)

Advanced Excel Course - More Functions

- Date and time functions
- Text functions
- Database functions
- Power Functions (CountIf, CountIFS, SumIF, SumIFS)

Advanced Excel Course - Formatting

- Using auto formatting option for worksheets
- Using conditional formatting option for rows, columns and cells

Advanced Excel Course - Macros

- Relative & Absolute Macros
- Editing Macrosofts

Advanced Excel Course - WhatIf Analysis

- Goal Seek
- Data Tables
- Scenario Manager

Advanced Excel Course - Charts

- Using Charts
- Formatting Charts
- Using 3D Graphs
- Using Bar and Line Chart together
- Using Secondary Axis in Graphs
- Sharing Charts with PowerPoint / MS Word, Dynamically
- (Data Modified in Excel, Chart would automatically get updated)

Advanced Excel Course - New Features Of Excel

- Sparklines, Inline Charts, data Charts
- Overview of all the new features

Advanced Excel Course - Final Assignment

- The Final Assignment would test contains questions to be solved at the end of the Course

VBA (VISUAL BASIC FOR APPLICATION) & MACROS

Create a Macro:

- Swap Values, Run Code from a Module, Macro Recorder, Use Relative References, FormulaR1C1, Add a Macro to the Toolbar, Macro Security, Protect Macro.

MsgBox:

- MsgBox Function, Input Box Function.

Workbook and Worksheet Object:

- Path and Full Name, Close and Open, Loop through Books and Sheets, Sales Calculator, Files in a Directory, Import Sheets, Programming Charts.

Range Object:

- Current Region, Dynamic Range, Resize, Entire Rows and Columns, Offset, From Active Cell to Last Entry, Union and Intersect, Test a Selection, Possible Football Matches, Font, Background Colors, Areas Collection, Compare Ranges.

Variables:

- Option Explicit, Variable Scope, Life of Variables.

If Then Statement:

- Logical Operators, Select Case, Tax Rates, Mod Operator, Prime Number Checker, Find Second Highest Value, Sum by Color, Delete Blank Cells.

Macro Errors:

- Debugging, Error Handling, Err Object, Interrupt a Macro, Macro Comments.

String Manipulation:

- Separate Strings, Reverse Strings, Convert to Proper Case, Count Words.

Date and Time:

- Compare Dates and Times, DateDif Function, Weekdays, Delay a Macro, Year Occurrences, Tasks on Schedule, Sort Birthdays.

Loop:

- Loop through Defined Range, Loop through Entire Column, Do Until Loop, Step Keyword, Create a Pattern, Sort Numbers, Randomly Sort Data, Remove Duplicates, Complex Calculations, Knapsack Problem.

Events:

- Before DoubleClick Event, Highlight Active Cell, Create a Footer Before Printing, Bills and Coins, Rolling Average Table.

Array:

- Dynamic Array, Array Function, Month Names, Size of an Array.

Function and Sub:

- User Defined Function, Custom Average Function, Volatile Functions, ByRef and ByVal.

Application Object:

- Status Bar, Read Data from Text File, Write Data to Text File.

ActiveX Controls:

- Text Box, List Box, Combo Box, Check Box, Option Buttons, Spin Button, Loan Calculator.

User form:

- User form and Ranges, Currency Converter, Progress Indicator, Multiple List Box Selections, Multicolumn Combo Box, Dependent Combo Boxes, Loop through Controls, Controls Collection, User form with Multiple Pages, Interactive User form.

Power BI

INTRODUCTION TO POWER BI

- Introduction to Business Intelligence (BI)
- Various BI tools
- Introduction to Power BI
- Why Power BI
- Power BI Components
- Introduction of Power BI Desktop
- Installation of Power BI Desktop

DATA VISUALIZATION

- Understanding Power View and Power Map
- Data visualization techniques
- Page layout & Formatting
- Power BI Desktop visualization
- Formatting and customizing visuals
- Column chart, Pie chart, Donut chart,
- Scatter chart, Funnel chart
- Include & exclude
- Geographical data visualization using Maps
- Drill down
- Drill through
- Page navigations
- Bookmarks
- Selection pane to show/hide visuals
- Comparing volume and value-based analytics
- Combinations charts (dual axis charts)
- Filter pane
- Slicers
- Use of Hierarchies in drill down analysis
- Sync slicers
- Tooltips & custom tooltips
- Tables & matrix
- Conditional formatting on visuals

POWER BI SERVICE, PUBLISHING & SHARING

- Introduction to Power BI Service
- Introduction of workspaces
- Dashboard
- Creating & Configuring Dashboards
- Dashboard theme
- Reports vs Dashboards
- Sharing reports & dashboards

DATA TRANSFORMATION – SHAPING & COMBINING DATA

- Shaping data using Power Query Editor
- Formatting data
- Transformation of data
- Understanding of Data types
- Naming conventions & best practices to consider
- Working with Parameters
- Merge Query
- Append Query
- Group by of data (aggregation of data)
- Duplicate & Reference tables
- Fill
- Pivot & Un-pivot of data

- Custom columns
- Conditional columns
- Replace data from the tables
- Split columns values
- Move columns & sorting of data
- Detect data type, count rows & reverse rows
- Promote rows as column headers
- Hierarchies in Power BI

DATA MODELING

- Introduction of relationships
- Creating relationships
- Cardinality
- Cross filter direction
- Use of inactive relationships

DAX

- Introduction of DAX
- Why DAX is used
- DAX syntax
- DAX functions
- Context in DAX
- Calculated columns using DAX
- Measures using DAX
- Calculated tables using DAX
- Learning about table, information, logical, text, iterator,
- Time intelligence functions (YTD, QTD, MTD)
- Cumulative values, calculated tables, and ranking and rank over groups
- Date and time functions



Institute of Ai

Reach us at:

info@drixcat.com | +91-9953356262

**Placement
Partners:**

